Simone **Colli**

Freelance Developer & Photographer | M.Sc. in Computer Science

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Education

University of Parma
Parma, Italy

Master's degree in Computer Science, Grade: 110/110 cum laude — *Sept. 2024 - July 2026*

Thesis title: Enhancing computational pangenomics workflows. Advisor Prof. V. Bonnici

University of Parma
Parma, Italy

Bachelor's degree in Computer Science, Grade: 108/110 — *Sept. 2021 - July 2024*

Thesis title: A parallel algorithm for computing sequence homology in pangenomic analysis. Advisor Prof. V. Bonnici

havce CTF team, University of Parma
Parma, Italy

Cybersecurity course — *2021 & 2024*

A cybersecurity training program for university and high school students organized by the Cybersecurity National Lab. (CyberChallenge.it)

Artistic roller skater
Italy

High-level athlete — *2007 - 2023*

I believe that sports, especially when practiced at a high level, can be considered a means of teaching; not purely under a professional perspective, but a personal one, because it teaches the values of organization, perseverance, patience, resilience, and many other skills that fall under the category of soft skills.

Experience

Self-employed
Remote

Freelance Software Developer — *January 2026 - Present*

- Designed and developed custom software solutions for private clients and small businesses, managing the full software development life cycle.
- Developed web applications and automation tools to streamline business processes.
- Managed client relationships, including requirement analysis, project estimation, and post-release maintenance.

Self-employed

Freelance Photographer — *January 2026 - Present*

Photography services for individuals and events. To handle photo delivery to clients, I independently designed and developed the *Photography delivery* platform (see the Projects section).

unipr-org

Co-founder & Maintainer — *January 2024 - Present*

An organization that brings together University of Parma students to collaborate on open-source educational materials and projects. (github.com/unipr-org)

Tutor
Parma, Italy

University of Parma; in-person & online — *February 2022 - June 2026*

- University tutor for the Bachelor's degree in Computer Science: Software Engineering (September 2024 - December 2024; September 2025 - December 2025) and Operating Systems (February 2025 - June 2025; February 2026 - June 2026).
- Freelance tutoring for university and high-school students. Topics: C, C++, Java, Python, algorithms and data structures, web development.

Seminars and workshops
Parma, Italy

University of Parma — *January 2025 - June 2025*

- Workshop "Tutoring from Different Perspectives and Supporting IT Tools" (January 24, 2025): presented and discussed student, tutor and lecturer perspectives and the challenges related to the tutoring role.
- Workshop "Use of Graphics Libraries for Software Implementation" (June 16-19, 2025): intensive bootcamp on Python fundamentals and graphics libraries, culminating in the development of the Pong game.

Internships

University of Parma — *February 2024 - September 2026*

- Bachelor's internship focused on the development of parallel software for pan-genomic analysis.
- Master's internship focused on enriching computational pangenomics workflows, including pipeline management, data visualization, file management and downstream analysis; this work later became the master's thesis.

Scientific publications and conferences

PanDelos-plus: A parallel algorithm for computing sequence homology in pangenomic analysis

PLOS Computational Biology, 22(9):e1014724 *2026*

Simone Colli, Emiliano Maresi, Vincenzo Bonnici. DOI: 10.1371/journal.pcbi.1014724

BBCC-Meetings *Napoli*

BBCC2024 *27-29 November 2024*

Oral presentation of PanDelos-plus.

Professional and personal projects

MGP – Production management

Work (continuously expanding)

- Full-stack web platform for preparing a manufacturing company's work orders, replacing a process spread across Excel files, Visual Basic macros and PowerShell scripts: the operator fills in the order and the system runs the calculations, generates the production documents and files (PDF, DXF, TLF, CSV) and returns an archive already organised by product type and destination.
- Customer-configurable rule engine: input fields, constants, formulas, conditions and checkpoints are defined in tables by administrators, and a dedicated microservice derives a dependency graph from them and propagates the calculations, so the company updates its own product models with no code change.
- Truck load planning (heuristic nesting) for both production and quotations, with drag & drop review and generated plan PDFs: preparing the plan by hand in AutoCAD took hours of work, the engine proposes an arrangement in 10–15 seconds.
- Polyglot microservices architecture documented with ADRs and C4 diagrams; OT/IT integration with factory PLCs over OPC-UA and constraint-based optimization with MiniZinc and Google OR-Tools. Stack: Laravel (PHP), React / TypeScript, Python (FastAPI/Flask), MySQL, Docker, Kubernetes (k3s).

Photography delivery

Personal (beyond prototype, in active testing)

- Self-hosted platform supporting freelance photography work: private galleries protected by an access code, cart, order management and full-resolution file delivery after payment.
- Stack: Python (FastAPI), Nuxt 3 / Vue, PostgreSQL, Redis, RQ (async workers), Docker.
- Strong focus on security and GDPR compliance (AES-256-GCM encryption of billing data, Argon2id, audit log) and a complete quality pipeline: multi-tool security scanning and strict type-checking.

Freelance hub

Personal (v1 complete)

- Self-hosted CRM and automation platform for freelance work, built to unify information previously spread across Obsidian folders, forms, email and WhatsApp: it covers the full cycle from lead to project and work done, through to quotations, contracts, payments and reporting.
- Business intelligence module with a revenue forecast at three, six and twelve months built on history, recurring income and the recent trend; analyses of revenue and margins, estimated against actual hours, income per hour and conversions by acquisition source.
- Workload planning: working hours, exceptions and fixed blocks (personal commitments included) feed an optimization service that proposes how to spread the tasks while respecting deadlines and constraints.
- Microservices with Gmail/Google Calendar/webhook integrations and an assistant on locally run LLMs (Ollama, LM Studio), so client data never leaves the machine. Engineered with TDD: 843 tests with a 100% coverage threshold enforced by the pipeline, strict type-checking (mypy/TypeScript) and a complete DevSecOps pipeline (SAST, SBOM, multi-tool security scanning, SonarQube). Stack: Python (FastAPI), GraphQL (Strawberry), Nuxt 3 / Vue, PostgreSQL, MiniZinc.

Personal website & portfolio

Personal

- Bilingual (IT/EN) portfolio site with case studies of the main projects: the process they replaced, the solution adopted, and technical diagrams drawn in SVG that follow the light/dark theme.
- Static generation: detail routes are pre-rendered from the project data, the sitemap is derived from those same routes, and SEO tags are injected into the `head` at build time (Open Graph, Twitter Card, canonical, Person structured data).
- Stack: React 19, TypeScript, Tailwind CSS 4, Vite; internationalization with i18next and continuous deployment to GitHub Pages via GitHub Actions.

University and outreach projects

Quantum Portfolio Optimization

Python *2025*

Development of a Python-based quantum portfolio optimization tool, leveraging the power of quantum computing to enhance portfolio management and investment strategies. The project explores the potential of hybrid algorithms VQE e QAOA, that combine classical and quantum computing, to solve Quadratic Unconstrained Binary Optimization (QUBO) problems. Simulations have been designed and implemented using Qiskit to evaluate algorithm performance under noiseless and noisy conditions.
(github.com/simonecolli/quantum-portfolio-optimization)

Ottimizzazioni garanti accademici

ASP, Python *2024*

Development of an automated system for assigning academic guarantors to university courses, adhering to complex ministerial regulations. The project includes the creation of a data preprocessing pipeline and the implementation of a system based on Answer Set Programming (ASP) for dynamic optimization of resource allocation. The system ensures scalability across various datasets, reduces reliance on external staff, and meets institutional and regulatory requirements. (github.com/simonecolli/ottimizzazione-garanti-accademici)

PanDelos-plus

C++, Python, Bash *May 2024 - September 2026*

A C++ based, parallelized revision of PanDelos (github.com/InfOmics/PanDelos) software for pan-genome content discovery. (github.com/simonecolli/PanDelos-plus)

Deep Neural Network Library

C++, Python *2024*

A flexible, modular, easy-to-use, and high-performance C++ library for building, training, and using deep neural networks. (github.com/unipr-org/deep-neural-network)

MazeProject

HTML, CSS, JS *2022 - 2025*

A user-friendly web-based application developed for the European Research's Night, aimed at making the understanding of graph-based search algorithms more interesting, easier, and enjoyable. (github.com/simonecolli/MazeProject)

Technical skills

Main computer languages	C++, Java, Python, TypeScript, PHP, SQL, Bash scripting, HTML, JavaScript, CSS, SASS, C, YAML, LaTeX.
Databases	PostgreSQL, MySQL, Redis.
Frameworks and other tools	React, Vue.js, Nuxt, Vite, Spring, Laravel, FastAPI, GraphQL, Tailwind CSS, Bootstrap, Docker, Kubernetes, Ansible, Grafana, git, GitHub, GitHub Actions, nginx, Apache Web Server, on-premise deployment, LLM & RAG, Jupyter Notebook.
Creative instruments	Microsoft Office, GIMP, Adobe Illustrator, Adobe Lightroom Classic, Adobe Photoshop, DaVinci Resolve, Audacity.
Soft skills	Organization, problem solving, teamwork, stress management, self-development, resilience, flexibility, patience, punctuality, perseverance.
Languages	Italian (First language), English (Intermediate).

These too, when needed

Other computer languages	C#, Objective-C, Markdown, StrictDoc, Matlab.
Operative systems	Linux, Windows, Windows Server 2019, MAC OSX.
Environments	Visual Studio Code, Eclipse, Android Studio, Xcode, Google Colab, MATLAB IDE, Sublime Text, Nano.
Other tools	Flask, jQuery, Hyper-V, Oracle Virtual Box, OpenMP, MPI, CUDA, Masterwood, AutoCAD, Adobe Premiere Pro, Darktable.